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Zhengchun Liu, PhD

Senior Applied Scientist

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Senior Applied Scientist with 10+ years of experience building large-scale machine learning and data infrastructure systems. Currently leading ML initiatives in Amazon Redshift focused on autonomous data warehouse optimization and AI-driven analytics. Previously a tenured computer scientist at Argonne National Laboratory leading $12M+ federally funded research programs in AI for Science and HPC systems. Expertise includes large-scale ML systems, distributed data infrastructure, query optimization, performance modeling, and agentic AI.

Leadership Highlights

- Technical lead for ML systems powering autonomous optimization in Amazon Redshift.
- Principal investigator of 10+ NSF/DOE projects totaling $12M+ funding.
- Led interdisciplinary teams spanning ML, HPC, and large-scale data infrastructure.

Technical Skills

Machine Learning systems, Predictive Modeling, Agentic AI; distributed data systems, HPC; Python, C++, SQL, CUDA.

Professional Experience

Amazon Web Services (AWS), AI Labs **Aug. 2022 — present**
Senior Applied Scientist

- Technical lead for machine learning systems powering autonomous optimization in Amazon Redshift (AWS data warehouse platform), enabling automated scaling, scheduling, and query optimization across large-scale production workloads.
- Developed ML models supporting **billions of optimization decisions per day** for resource management and performance optimization in cloud data warehouse infrastructure.
- Lead scientist developing **agentic AI systems for automatic analytical SQL generation**, enabling natural-language-driven analytics workflows for Amazon Ads.

Argonne National Laboratory **Aug. 2016 — Aug. 2024**
Postdoctoral(2016/08), Assistant Computer Scientist (2019/08), Tenured Computer Scientist (2022/05), Sabbatical leave (2022/08)

- Principal Investigator or co-PI on 10+ NSF/DOE funded projects totaling $12M+, focused on AI-for-Science and High-Performance Computing systems.
- Led research group developing **AI and HPC systems for scientific computing workloads** on leadership-class supercomputers.
- Lead Author of **50+** peer-reviewed papers (5 **best paper awards**) in data infrastructure, distributed systems, and ML systems.
- Developed data and ML infrastructure for large-scale scientific data processing; project won the Technology Challenge at ACM/IEEE Supercomputing (SC'19).

University of Chicago **Mar. 2018 — Aug. 2019**
Research Scientist

- Led research on architecture of autonomous scientific systems combining large-scale data infrastructure and machine learning.
- Developed performance modeling and system characterization techniques for scientific applications on supercomputers.

Embedded Systems Consulting (Founder) **Sep. 2010 — Nov. 2013**

- Founded and operated embedded systems consulting business delivering 10+ engineering contracts.
- Developed board support packages for µC/OS-II real-time operating system.
- Designed and developed streaming systems for vehicle event data recorder devices (connected IoT).

Education

- **PhD., Computer Science**, Autonomous University of Barcelona, Spain — Jul. 2016
  - `With Honors(Cum Laude); Extraordinary Doctorate Award(top thesis distinction).`
- **MSc., Guidance, Navigation and Control**, Northwestern Polytechnical University, China — Apr. 2013
- **BSc., Aircraft Manufacturing Engineering**, Northwestern Polytechnical University, China — Jun. 2010
  - `National Scholarship` (top 0.2%) ×2, 2008 and 2009.

Professional Service

- **Editorship**: Journal of Future Generation Computer Systems (FGCS), Impact Factor: 7.2. — 2020 – 2023
- **Workshop Co-Chair**: SRMPDS 2017, 2018, 2019, 2020; AI-Science'19.
- **Technical Program Committee**: ICDS'19; DAAC'17-19; DLS'19; ICDCS'20; SC'20; HPCC'20; HiPC'22; SC'23; IPDPS'23,24.

Selected Publications

* denotes student/post-doc supervised. **Full list**: Google Scholar

Papers on Machine Learning and Computer Systems

1. Zhou, Y., Liu, C., Urgaonkar, B., Wang, Z., Mueller, M., Zhang, C., Zhang, S., Pfeil, P., Horn, D., **Liu, Z.** and Pagano, D., ***PBench: Workload Synthesizer with Real Statistics for Cloud Analytics Benchmarking***. VLDB'25.
2. Alexander van Renen, Dominik Horn, Pascal Pfeil, Kapil Eknath Vaidya, Wenjian Dong, Murali Narayanaswamy, **Zhengchun Liu**, Gaurav Saxena, Andreas Kipf, Tim Kraska. ***Why TPC is not enough: An analysis of the Amazon Redshift fleet***. VLDB'24.
3. Vikram Nathan, Vikramank Singh, **Zhengchun Liu**, Mohammad Rahman, Andreas Kipf, Dominik Horn, Davide Pagano, Gaurav Saxena, Balakrishnan Narayanaswamy, Tim Kraska. ***Intelligent Scaling in Amazon Redshif***. SIGMOD'24.
4. Ziniu Wu*, Ryan Marcus, **Zhengchun Liu**, Parimarjan Negi, Vikram Nathan, Pascal Pfeil, Gaurav Saxena, Mohammad Rahman, Balakrishnan Narayanaswamy, Tim Kraska. ***Stage: Query Execution Time Prediction in Amazon Redshift***. SIGMOD'24.
5. **Zhengchun Liu**, Rajkumar Kettimuthu, Michael E. Papka, Ian Foster. ***FreeTrain: A Framework to Utilize Unused Supercomputer Nodes for Training Neural Networks***. IEEE/ACM International Symposium in Cluster, Cloud, and Grid Computing (CCGrid'23).
6. Ahsan Ali*, Hemant Sharma, Rajkumar Kettimuthu, Peter Kenesei, Dennis Trujillo, Antonino Miceli, Ian Foster, Ryan Coffee, Jana Thayer, **Zhengchun Liu**. ***fairDMS: Rapid Model Training by Data and Model Reuse***. 2022 IEEE International Conference on Cluster Computing.
7. **Zhengchun Liu**, Ahsan Ali*, Peter Kenesei, Antonino Miceli, Hemant Sharma, Nicholas Schwarz, Dennis Trujillo, Hyunseung Yoo, Ryan Coffee, Ryan Herbst, Jana Thayer, Chun Hong Yoon, Ian Foster. ***Bridge Data Center AI Systems with Edge Computing for Actionable Information Retrieval***. XLOOP@SC'21. ***Best Paper Awarded***.
8. Tirthak Patel, **Zhengchun Liu**, Rajkumar Kettimuthu, Paul Rich, Bill Allcock, Devesh Tiwari. ***Job Characteristics on Large-Scale Systems: Long-Term Analysis, Quantification, and Implications***. The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'20).
9. **Zhengchun Liu**, Ryan Lewis*, Rajkumar Kettimuthu, Kevin Harms, Philip Carns, Nageswara Rao, Ian Foster and Michael Papka. ***Characterization and Identification of HPC Applications at Leadership Computing Facility***. International Conference on Supercomputing (ICS'20).
10. Yuanlai Liu*, **Zhengchun Liu**, Rajkumar Kettimuthu, Nageswara Rao, Zizhong Chen and Ian Foster. ***Data transfer between scientific facilities - bottleneck analysis, insights and optimizations***. IEEE/ACM International Symposium in Cluster, Cloud, and Grid Computing (CCGrid'19).
11. Rajkumar Kettimuthu, **Zhengchun Liu**, David Wheeler, Ian Foster, Katrin Heitmann, Franck Cappello. ***Transferring a Petabyte in a Day***. Future Generation Computer Systems, 2018(88).
12. **Zhengchun Liu**, Rajkumar Kettimuthu, Ian Foster and Nageswara S.V. Rao. ***Cross-geography Scientific Data Transfer Trends and User Behavior Patterns***. International Symposium on High-Performance Parallel and Distributed Computing (HPDC'18).
13. **Zhengchun Liu**, Prasanna Balaprakash, Rajkumar Kettimuthu and Ian Foster. ***Explaining Wide Area Data Transfer Performance***. International Symposium on High-Performance Parallel and Distributed Computing (HPDC'17), 167-178.

Papers on HPC and AI for Science

1. Anakha V Babu, Tao Zhou, Saugat Kandel, Tekin Bicer, **Zhengchun Liu**, William Judge, Daniel J Ching, Yi Jiang, Sinisa Veseli, Steven Henke, Ryan Chard, Yudong Yao, Ekaterina Sirazitdinova, Geetika Gupta, Martin V Holt, Ian T Foster, Antonino Miceli and Mathew J Cherukara. ***Deep Learning at the Edge Enables Real-time Streaming Ptychographic Imaging***. Nature Communications 14, 7059 (2023).
2. **Zhengchun Liu**, Hemant Sharma, Jun-Sang Park, Peter Kenesei, Antonino Miceli, Jonathan Almer, Rajkumar Kettimuthu and Ian Foster. ***BraggNN: Fast X-ray Bragg Peak Analysis Using Deep Learning***. IUCrJ, Vol. 9, No. 1, 2022.
3. Aniket Tekawade, **Zhengchun Liu**, Peter Kenesei, Tekin Bicer, Francesco De Carlo, Rajkumar Kettimuthu, Ian Foster. ***3D Autoencoders For Feature Extraction In X-ray Tomography***. 2021 IEEE International Conference on Image Processing.
4. Jiali Wang, **Zhengchun Liu**, Ian Foster, Won Chang, Rajkumar Kettimuthu, Rao Kotamarthi. ***Fast and accurate learned multiresolution dynamical downscaling for precipitation***. journal of Geoscientific Model Development.
5. **Zhengchun Liu**, Tekin Bicer, Rajkumar Kettimuthu, Doga Gursoy, Francesco De Carlo and Ian Foster. ***TomoGAN: Low-Dose Synchrotron X-Ray Tomography with Generative Adversarial Networks***. Optical Society of America A, 2020, Vol. 37, No. 2.
6. **Zhengchun Liu**, Tekin Bicer, Rajkumar Kettimuthu and Ian Foster. ***Deep Learning Accelerated Light Source Experiments***. IEEE/ACM Deep Learning on Supercomputers DLS@SC'19.